

Composition Order and Inverse Domains

Answer Key

Algebra 2 — Error Analysis

Quick Answers

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|-------|-------|--------|----------|
| 1. 15 | 3. 5 | 5. x | 7. 3, -3 |
| 2. 2 | 4. -5 | 6. 7 | 8. — |
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Common wrong answers

4. If they got 7: composed in the reverse order, computing $g(f(4))$ instead of $f(g(4))$
6. If they got -1: took the minus branch of the square root, which lands outside the domain
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1. $f(x) = 2x - 5$, $g(x) = x^2 + 1$; find $(f \circ g)(3)$.

Step 1. $(f \circ g)(3)$ means $f(g(3))$ — the function written on the *right* acts first.

Step 2. $g(3) = 3^2 + 1 = 10$.

Step 3. $f(10) = 2(10) - 5 = 15$.

15

2. Same f and g ; find $(g \circ f)(3)$.

Step 1. Now f acts first: $(g \circ f)(3) = g(f(3))$.

Step 2. $f(3) = 2(3) - 5 = 1$.

Step 3. $g(1) = 1^2 + 1 = 2$.

Comparison to state: $15 \neq 2$, so composition is *not* commutative — swapping the order changes the answer, and one counterexample settles it.

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3. $f(x) = 3x + 7$; find $f^{-1}(22)$.

Step 1. $f^{-1}(22)$ is the input whose output is 22, so solve $3x + 7 = 22$ rather than substituting 22 into f .

Step 2. $3x = 15$, so $x = 5$.

Step 3. Check forward: $f(5) = 15 + 7 = 22$ ✓

$x = 5$

4. Find and fix. $f(x) = 3x + 1$, $g(x) = x - 6$; $(f \circ g)(4)$.

Step 1. Nia's first move was $f(4) = 13$, then she subtracted 6. That is $g(f(4)) = (g \circ f)(4) = 7$ — the reverse composition.

Step 2. The correct order feeds 4 to g first: $g(4) = 4 - 6 = -2$.

Step 3. Then $f(-2) = 3(-2) + 1 = -5$.

One-sentence diagnosis to accept: “She applied f first; in $f \circ g$ the inner function g is applied first.”

-5

5. $f(x) = 3x + 7$, $f^{-1}(x) = \frac{x-7}{3}$; show $f(f^{-1}(x)) = x$.

Step 1. Substitute the whole inverse expression into f : $f(f^{-1}(x)) = 3 \cdot \frac{x-7}{3} + 7$.

Step 2. The 3 and the denominator 3 cancel: $(x-7) + 7$.

Step 3. $x - 7 + 7 = x$ for every real x , which is exactly the identity that certifies an inverse.

$$\boxed{f(f^{-1}(x)) = x}$$

6. Find and fix. $h(x) = (x-3)^2$ on $x \geq 3$; Sam's $h^{-1}(x) = 3 - \sqrt{x}$ gives $h^{-1}(16) = -1$.

Step 1. An inverse must return values *inside the domain of h* , and $-1 \geq 3$ is false, so Sam's output cannot be an input of h .

Step 2. Build the rule honestly: from $y = (x-3)^2$, $\sqrt{y} = |x-3| = x-3$ because $x \geq 3$ makes $x-3$ non-negative. So $h^{-1}(x) = 3 + \sqrt{x}$ — the plus branch is forced by the restriction.

Step 3. $h^{-1}(16) = 3 + 4 = 7$, and checking forward, $h(7) = (7-3)^2 = 16$ ✓

Diagnosis to accept: “He kept the minus branch of the square root, which leaves the restricted domain $x \geq 3$.”

7

7. $g(x) = x^2$ on all of $\mathbb{R}$; solve $x^2 = 9$.

Step 1. $x^2 - 9 = 0$ factors as $(x-3)(x+3) = 0$, so the equation has two real solutions, not one.

Step 2. $x = 3$ and $x = -3$ both square to 9.

What it proves: two different inputs share the output 9, so g is not one-to-one on $\mathbb{R}$ and has no inverse there. Writing $g^{-1}(x) = \sqrt{x}$ silently keeps only the $x \geq 0$ half of the domain; the claim is legitimate only after g is restricted to $x \geq 0$.

$$\boxed{x = 3 \text{ or } x = -3}$$

8. $f(x) = \sqrt{x-1}$, $g(x) = x^2 + 1$; Marcus's claim.

Step 1. The step $\sqrt{x^2} = x$ is the flaw: $\sqrt{x^2} = |x|$, which equals x only when $x \geq 0$. For $x = -4$, $f(g(-4)) = \sqrt{17-1} = 4$, not -4 .

Step 2. He checked only one direction. The second check, $g(f(x)) = (\sqrt{x-1})^2 + 1 = x$, needs $x-1 \geq 0$, i.e. $x \geq 1$ — and that is the restriction his one-sided check hid.

Step 3. Domains that make the pair genuine inverses: f has domain $x \geq 1$ with range $y \geq 0$; g must be restricted to $x \geq 0$, where its range is $y \geq 1$. Each function's domain is the other's range, which is the structural signature of an inverse pair.

Model answer (accept any equivalent wording): “ $\sqrt{x^2} = |x|$, so $f(g(x)) = x$ only for $x \geq 0$. He never checked $g(f(x))$, which forces $x \geq 1$. With $f : x \geq 1 \rightarrow y \geq 0$ and $g : x \geq 0 \rightarrow y \geq 1$ the two are genuine inverses.”